

Evaluation

Website Design

The project is called Be Mine Forever, which is a wedding planner website designed to help couples organise their wedding more easily online.

A few years ago, I originally started this idea as a mobile application concept. Now I have evolved the idea further and developed it into a responsive website that works across different screen sizes and devices.

The website allows couples to:

- View the Home page
- Create a new account
- Log in and log out
- Create and edit a wedding plan
- Tick completed wedding tasks
- View a wedding date countdown
- Monitor their budget using a budget counter
- Browse vendors by category
- Create a guest list
- View the guest list

Accessibility was also considered during development by implementing keyboard navigation to make the website easier to use for different users.

Development

During development, I first created the ERD (Entity Relationship Diagram) to plan the database structure properly before creating the system itself.

I then created the database tables while considering:

- Relationships between tables
- Constraints

After doing the database, I developed the API backend for the main endpoints using PHP and MySQL.

Injection Prevention

To prevent injection attacks, I properly validated user inputs before they are processed by the system, so that attackers cannot write malicious code in the login form to bypass authentication.

I also used prepared statements instead of directly inserting user input into SQL queries. This ensured that user input was treated as data rather than executable SQL commands.

API Validations and HTTP Responses

The API was designed to return proper HTTP response codes to improve validation and error handling, including:

- 200 – OK
- 201 – Created
- 400 – Bad Request
- 401 – Unauthorised
- 403 – Forbidden
- 404 – Not Found
- 405 – Method Not Allowed
- 409 – Conflict
- 500 – Server Error

These responses helped with both debugging and frontend integration during development.

Broken Access Control

Role-based access control was implemented within the API to restrict users based on their role permissions.

Administrators can:

- Create vendors, categories, tasks, and roles
- Delete vendors and categories

Couples can:

- Create an account
- Create wedding plans and guest lists, etc.
- View categories, wedding plans, and guest information
- Update wedding dates, guest counts, budgets, and tasks

API Testing

I downloaded and used Postman to test the API endpoints throughout development.

Testing included:

- Input validations
- Error handling
- JSON responses
- Different HTTP methods such as:
 - POST
 - GET
 - PATCH
 - PUT
 - DELETE

Postman also helped organise testing by grouping related endpoints, making testing more efficient and structured.

Exportation of Results

Exportation of results mainly involved transferring data from the API to the frontend using JSON responses.

The website relied on cURL requests to communicate with the API endpoints and fetch data such as:

- Wedding plans
- Vendor categories
- Guest information

JSON was used because it is lightweight, structured, and easy for the frontend to process dynamically. cURL allowed the frontend pages to send requests to the API and display the returned data correctly within the website interface.

Future Improvements

Authentication Improvements

To improve authentication security further, I plan to implement additional features such as:

- Verification codes when users forget their password
- Strong password suggestions
- More secure session handling

This would help reduce risks such as password guessing and session hijacking.

SEO Improvements

During development, I improved the URL structure for categories by using slugs instead of numeric IDs. Example:

http://localhost/BeMine_wedding_website/vendors.php?category=florists

instead of:

http://localhost/BeMine_wedding_website/vendors.php?category=1

This made the URLs easier for users to read, and it also improved Search Engine Optimisation because search engines can recognise keywords such as “florists”.

In future development, this approach could be added to other sections of the website.

Accessibility and Performance

To improve the website further, I also plan to:

- Add ALT text for all images to improve accessibility and SEO
- Implement caching for frequently used data, such as vendor categories

- Improve performance optimisation for larger amounts of data and future scalability